

AROB 2025

General introduction

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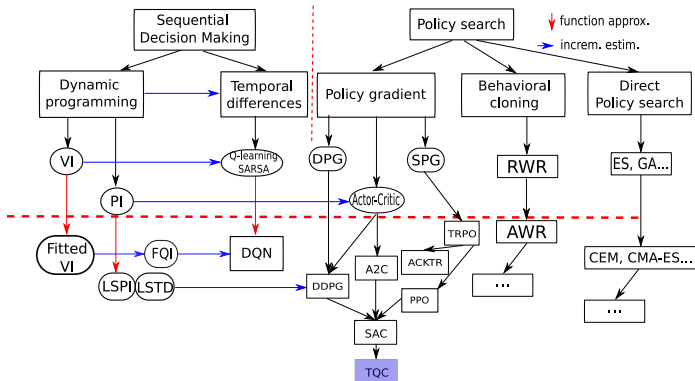
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Content

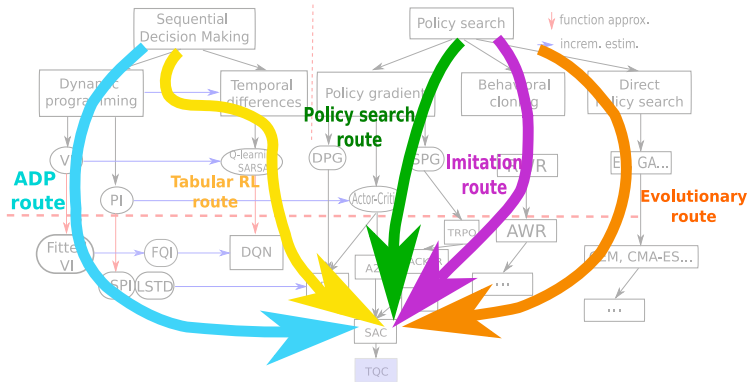
1. Regression
2. Model-free reinforcement learning (tabular)
3. Statistics in RL, hyper-parameter tuning
4. Model-based reinforcement learning (tabular)
5. Evolution strategies (ES, CEM, CMA-ES)
6. Evolutionary robotics
7. Collective evolutionary robotics
8. Multi-objective evolutionary optimization
9. Evolutionary optimization of structures
10. Diversity seeking methods (novelty search, QD)

RL part: The Big Picture



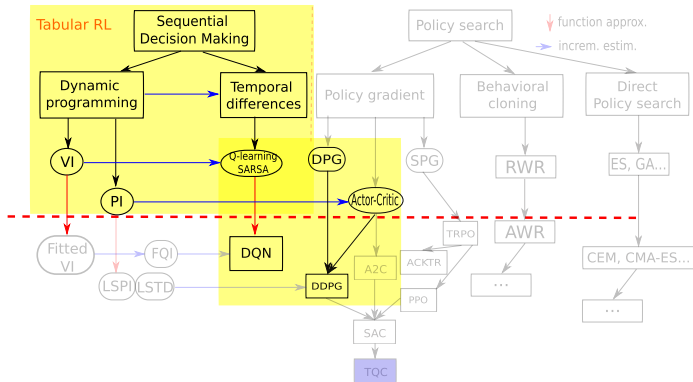
- A very partial view of the whole RL literature

The five routes to deep RL



- Five different ways to come to Deep RL

The Tabular RL route



- ▶ The favorite route of beginners
- ▶ Start from Sutton&Barto, present Q-learning, SARSA and Actor-Critic
- ▶ Add function approximation with NNs, go to DQN, then DDPG and TD3 (in M2)

Organization (my part)

- ▶ Lessons about algorithms, concepts and tools
- ▶ During lessons: rehearsal questions about the concepts
- ▶ Labs: coding algorithms, practicing, tuning hyper-parameters (notebooks)
- ▶ Two small projects
- ▶ Lessons are available in video
- ▶ Everything available in advance, supplementary material...

Evaluation

- ▶ One lab per week, average over 10 labs
- ▶ Two small projects inserted in my part
- ▶ Questions about lessons are not evaluated
- ▶ Delay is penalized (1 point over 5 per day for any deadline missed)
- ▶ Technical details on the Moodle page
- ▶ Advices:
 - ▶ Watch lessons in advance, ask questions in classroom
 - ▶ Anticipate for small projects
- ▶ Look at the Moodle page for more details

Any question?



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Sutton, R. S. and Barto, A. G. (1998).
Reinforcement Learning: An Introduction.
MIT Press.